

Slope of Linear Functions from Equations

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The **slope** of a linear function describes its **rate of change**.

Slope-Intercept Form

In the equation $y = mx + b$, the coefficient m is the **slope**.

Positive and Negative Slope

A **positive slope** means the graph rises from left to right.

A **negative slope** means the graph falls from left to right.

Zero Slope

A slope of 0 indicates a **horizontal line**.

Steepness

The greater the absolute value of the slope, the **steeper** the line.

Rewriting Equations

Equations may need to be rewritten in the form $y = mx + b$ before identifying the slope.

Interpreting Slope

The slope can be interpreted as the amount the output changes for every 1 unit increase in the input.

Practice Exercises

Practice 1–5

Determine the **slope** of each equation.

$$y = 6x + 2$$

$$y = -4x + 9$$

$$y = \frac{3}{2}x - 5$$

$$y = -\frac{5}{3}x + 1$$

$$2y = 8x - 10$$

Practice 6–10

Determine the **slope** of each equation.

$$3y = -12x + 15$$

$$x + y = 7$$

$$4x + 2y = 14$$

$$5x - y = 11$$

$$6x + 3y = -9$$

Activities

Activity 1–5

Determine the **slope** of each equation.

$$y = 8x - 3$$

$$y = -7x + 4$$

$$y = \frac{5}{4}x + 2$$

$$y = -\frac{7}{2}x - 6$$

$$2y = 14x + 8$$

Activity 6–10

Determine the **slope** of each equation.

$$5y = -20x + 25$$

$$2x + y = 10$$

$$6x + 3y = 18$$

$$7x - y = 13$$

$$8x + 4y = -16$$

Thank you!